

R2

Below is a table of the results from the different linear classifiers.

This is for the **Diabetes** dataset

Model	Cross Validated	Test Weighted
Perceptron before RBF	0.686	0.596
Perceptron after RBF	0.651	0.531
Logistic regression before RBF	0.737	0.693
Logistic regression after RBF	0.627	0.735
Multilayer perceptron	0.775	0.731

Observations:

- For the perceptron, RBF did not improve the perceptron performance
- For logistic regression, LR before kernel had the strongest cross validated performance and after the kernel, the cross validation performance decreased but F1 scores slightly increased
- MLP slightly outperformed logistic regression on cross validation

Conclusion:

The diabetes dataset seems to be able to be easily linearly separated after normalization and imputation. The RBF kernel did not improve data enough to note which means that the dataset does not need nonlinear transformation. Logistic regression seems to perform the best out of all of the linear models and MLP did a good job as well because it can model mild nonlinear relationships.

This is for the **Glass** dataset

Model	Cross Validated	Test Weighted
Perceptron before RBF	0.410	0.584
Perceptron after RBF	0.426	0.473
Logistic regression after RBF default gamma	0.193	0.334
Logistic regression after gamma tuning	0.588	
Multilayer perceptron	0.674	0.724

Observations:

- For the perceptron, the performance seems to be relatively unstable.
- For logistic regression, RBF tuning seemed to improve cross validation performance, but test performance is still not very good.
- For MLP, it obviously did very well compared to the other 2 models

Conclusion:

Since the glass dataset has multiple classes and a lot of overlap, there isn't a lot of linear structure so the linear models struggled a lot. The RBF kernel seemed to help a bit, but not by much. It did improve more once the gamma was tuned though. With the way that MLP obviously did a much better job, it confirms that the glass dataset very much had nonlinear class boundaries.

Overall Conclusion:

The best overall model for both of the datasets was MLP. The improvements with MLP were small for the diabetes dataset and were substantial for the glass dataset. This shows that the diabetes dataset was closer to linearly separable while the glass one had a more nonlinear structure.

R3

The virtualization shows that the original feature space has a lot of class overlap, especially in the glass dataset. The linear decision boundary cuts across regions where multiple classes mix which can help explain the poor performance of the linear classifiers. The RBF kernel transforms the feature space into a higher dimensional representation where the classes are more separable, which improves performance for the nonlinear datasets.

R4

The method that I used to determine the 5 least significant predictors is to:

1. Train logistic regression
2. Extract coefficients
3. Take the absolute value of said coefficients
4. Average them across classes
5. Rank the features by magnitude

I chose this method because in logistic regression coefficient magnitude reflects the feature contribution to decision boundary.

Diabetes Results

Top predictors

- Glucose
- BMI
- Pregnancies

- Diabetes pedigree function
- Skin thickness

Bottom 5 predictors

- Diabetes pedigree function
- Skin thickness
- Blood pressure
- Age
- Insulin

There are only 8 predictors, so there is some overlap.

After dropping the least 5 significant F1 is approximately 0.717. Before it was 0.693

This means that removing the weakest predictors did improve the performance slightly, which means that there was some noise reduction.

Glass Results

Top predictors

- Al
- Na
- Mg
- Ba
- Si

Bottom 5 predictors

- Si
- Ri
- K
- Ca
- Fe

There are only 9 predictors so there is some overlap.

After dropping the least 5 significant F1 is approximately 0.546, which is a drop from the original.

The glass dataset relies on a lot of interacting features. Dropping the predictors reduces the model capacity and hurts its performance.

R5

The RBF kernel maps the data into a higher dimensional feature space where similarity between points is important. In this space, classes that are not linearly separable in the original space may become separable by a linear boundary. This would explain the improved performance that I saw after the kernel tuning, especially for the glass dataset.